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| United States Patent Application Publication | 20250276377 |
| Kind Code | A1 |
| Publication Date | September 04, 2025 |
| Inventor(s) | Nelson; Scott et al. |


## ADAPTIVE MACHINING USING BUILD SURFACE TOPOLOGY FOR ADDITIVE MANUFACTURING SYSTEMS


### Abstract

An additive manufacturing system includes an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component, a powder delivery device configured to direct a powder stream toward the melt pool, a machining device configured to machine the build surface, at least one topology sensor configured to generate topological data representative of a topology of the build surface, and a computing device configured to receive the topological data from the at least one topology sensor for a plurality of layers, identify differences between the topological data and specification data representative of a set of tolerances of the build surface, control the energy delivery device and the powder delivery device based on a set of deposition parameters, and control the machining device to machine the build surface based on the identified differences.



**Inventors:** **Nelson; Scott (Indianapolis, IN), Puhl; David James (Indianapolis, IN), Grafton-Reed; Clive (London, GB), Daum; Peter E. (Indianapolis, IN), Proctor; Robert F. (Indianapolis, IN), Heason; Christopher P. (London, GB)**

**Applicant:** **Rolls-Royce Corporation** (Indianapolis, IN); **Rolls-Royce plc** (London, GB)



**Family ID:** **96881598**

**Appl. No.:** **18/593672**

**Filed:** **March 01, 2024**


### Publication Classification

**Int. Cl.:** **B22F10/85** (20210101); **B22F10/28** (20210101); **B22F10/66** (20210101); **B33Y10/00** (20150101); **B33Y30/00** (20150101); **B33Y50/02** (20150101)

**U.S. Cl.:**

CPC **B22F10/85** (20210101); **B22F10/28** (20210101); **B22F10/66** (20210101); B33Y10/00 (20141201); B33Y30/00 (20141201); B33Y50/02 (20141201)

## Background/Summary

TECHNICAL FIELD

[0001] The disclosure relates to additive manufacturing techniques.

BACKGROUND

[0002] Additive manufacturing generates three-dimensional structures through addition of material layer-by-layer or volume-by-volume to form the structure, rather than removing material from an existing component to generate the three-dimensional structure. Additive manufacturing may be advantageous in many situations, such as rapid prototyping, forming components with complex three-dimensional structures, or the like. In some examples, additive manufacturing may utilize powdered materials and may melt or sinter the powdered material together in predetermined shapes to form the three-dimensional structures. In addition to initial fabrication, additive manufacturing techniques may include subsequent machining processes to finish a component.

SUMMARY

[0003] The disclosure describes additive manufacturing systems, and methods for operating additive manufacturing systems, that adaptively manufacture a component using topological data of a build surface. The topological data is obtained during initial fabrication of the component and provides information about the component that may not be available from topological scans of an as-fabricated component, including differences between target and as-deposited parameters of the build surface for various intermediate layers. The topological data may be used to control machining after deposition of the particular layer or after initial fabrication of the component, as the identified differences between the target and as-deposited parameters may indicate material or structural properties for which different machining parameters may be warranted. For example, topological variation in a section of a component may indicate differences in build quality, and correspondingly inform material removal parameters that account for the differences in build quality, such as lower (e.g., reduced) material removal rate for those particular sections. In this way, the additive manufacturing systems described herein may enable more accurate finishing of a fabricated component.

[0004] In some examples, the disclosure describes an additive manufacturing system that includes an energy delivery device, a powder delivery device, a machining device, one or more sensors, and a computing device. The energy delivery device is configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component. The powder delivery device is configured to direct a powder stream toward the melt pool. The machining device is configured to machine the build surface. The one or more sensors include at least one topology sensor configured to generate topological data representative of a topology of the build surface. The computing device is configured to receive the topological data from the at least one topology sensor for a plurality of layers and identify differences between the topological data and specification data representative of a set of tolerances of the build surface. The computing device is further configured to control the energy delivery device and the powder delivery device based on a set of deposition parameters, and control the machining device to machine the build surface based on the identified differences.

[0005] In some examples, the disclosure describes a method that includes receiving, by one or more computing devices, topological data from one or more sensors of an additive manufacturing system. The additive manufacturing system includes an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of a

component, a powder delivery device configured to direct a powder stream toward the melt pool, a machining device configured to machine the build surface, and the one or more sensors. The one or more sensors include at least one topology sensor configured to generate the topological data representative of the topology of the build surface. The method further includes identifying, by the one or more computing devices, differences between the topological data and specification data representative of a set of tolerances of the build surface. The method further includes controlling, by the one or more computing devices, the energy delivery device and the powder delivery device based on a set of deposition parameters, and controlling, by the one or more computing devices, the machining device to machine the build surface based on the identified differences.
[0006] The details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

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## Description

BRIEF DESCRIPTION OF DRAWINGS
[0007] FIG. **1** is a conceptual block diagram illustrating topology monitoring aspects of an example additive manufacturing system that includes a topology sensor configured to monitor topology of a build surface within the additive manufacturing system during an additive manufacturing technique.
[0008] FIG. **2** is a process flow diagram illustrating a mass flux and heat flux monitoring and control technique.
[0009] FIG. **3**A is a flowchart illustrating an example method for fabricating a component using topological data.
[0010] FIG. **3**B is a flowchart illustrating an example method for generating machining data using topological data.
[0011] FIG. **3**C is a cross-sectional side view diagram illustrating an intermediate build of a component that includes topological defects.
[0012] FIG. **3**D is a cross-sectional side view diagram illustrating identification of topological defects in an intermediate build of a component.
[0013] FIG. **3**E is a cross-sectional side view diagram illustrating generation of machining data that accounts for topological defects.
[0014] FIG. **4**A is a flowchart illustrating an example method for fabricating a component using topological data.
[0015] FIG. **4**B is a flowchart illustrating an example method for generating machining data using topological data.
[0016] FIG. **4**C is a cross-sectional side view diagram illustrating an intermediate build of a component that includes topological defects.
[0017] FIG. **4**D is a cross-sectional side view diagram illustrating identification of topological defects in an intermediate build of a component.
[0018] FIG. **4**E is a cross-sectional side view diagram illustrating machining of topological defects.
[0019] FIG. **4**F is a cross-sectional side view diagram illustrating deposition of an overlying layer on the machined layer.
[0020] FIG. **5**A is a flowchart illustrating an example method for fabricating a component using topological data.
[0021] FIG. **5**B is a flowchart illustrating an example method for generating adjusted deposition parameters using topological data.
[0022] FIG. **5**C is a cross-sectional side view diagram illustrating an intermediate build of a component that includes topological defects.

[0023] FIG. **5**D is a cross-sectional side view diagram illustrating identification of topological defects in an intermediate build of a component.

[0024] FIG. **5**E is a cross-sectional side view diagram illustrating deposition of a subsequent layer that accounts for the topological defects of the underlying layer.

[0025] FIG. **6**A is a conceptual diagram illustrating an example machine learning model.

[0026] FIG. **6**B is a conceptual diagram illustrating an example training process for a machine learning model.

## DETAILED DESCRIPTION

[0027] The disclosure generally describes techniques and systems for monitoring topology of a component during a blown powder additive manufacturing technique, such as a directed energy deposition (DED) technique, and using the measured topology in a subsequent machining technique, such as milling. During blown powder additive manufacturing, a component is built up by adding material to the component in sequential layers, such that the final component is composed of a plurality of layers of material. In blown powder additive manufacturing techniques for forming components from metals or alloys, an energy source may direct energy at a substrate to form a melt pool, and a powder delivery device may deliver a powder to the melt pool. At least some of the powder at least partially melts and is joined to the melt pool and, thus, the substrate. Once the initial fabrication is complete, various surfaces of the component may be further finished through machining to bring the component to desired dimensions and/or provide the component with a desired surface finish.

[0028] The properties of the final component, including the presence or absence of material defects and the resulting microstructure, are a function of a number of variables related to mass flux and heat flux. As such, measurement of mass flux and heat flux within the blown powder additive manufacturing system may enable characterization or prediction of final component properties, control of the blown powder additive manufacturing technique, quality assurance for the final component, and the like. Mass flux and heat flux measurements may relate to parameters of powder flow rate and melt pool size and temperature, respectively, and may not provide an accurate characterization of an as-deposited layer. For example, while mass flow may be measured along various points in a flow path of the powder, turbulence and other conditions near a deposition surface may cause variation in a deposition rate at a particular portion of a layer.

[0029] Due to inherent limitations of these and optical measurement techniques, material defects may not be detected by mass flux or heat flux measurements, and may not be detectable by external surface scans of the final component. However, such material defects may impact various mechanical properties, such as hardness and brittleness, that influence how material is removed from the component during finishing. For example, a portion of a material that includes various defects that result in higher brittleness may be machined differently than a portion of a material that is relatively defect-free. As a result, without information as to how the various intermediates layers were formed, subsequent machining steps may either undercompensate for the material defects by machining all surfaces of the component at less sensitive parameters (e.g., higher material removal rate) or overcompensate for the material defects by machining all surfaces of the component at more sensitive parameters (e.g., lower material removal rate).

[0030] In accordance with techniques of this disclosure, an additive manufacturing system may include a topology sensor for measuring a topology of material of the build surface added to the melt pool and a computing device that adaptively controls machining based on the topology of the build surface. The topology of the build surface may provide additional and/or alternative information for determining the properties of the final component, including defects and variations in microstructure of intermediate layers. For example, a localized difference in height within a layer may indicate a material defect or other change in microstructure that is actionable during subsequent machining, either immediately after deposition of the particular layer or after fabrication of the final component. The topology sensor may output topological data to the computing device, which analyzes the topological data and generates a model that represents the as-fabricated component, including defects in the final component indicated by the topological data. The computing device may further control a machining device to machine the as-fabricated component based on the model in a manner that accounts for the defects of the as-fabricated component indicated by the topological data. For example, the computing device may select machining parameters for machining portions of the component that include sub-surface defects that better preserve the integrity of the component. By monitoring the topology of the build surface of various layers during fabrication of the component, the systems described herein may enable a more complete representation of the final properties of the component for subsequent machining, and accordingly, more effectively and/or efficiently control operation of the machining device.

[0031] FIG. **1** is a conceptual block diagram illustrating an example manufacturing system **50** that includes an additive manufacturing system **10** configured to monitor a topology of a component **22** during an additive manufacturing technique. In the example illustrated in FIG. **1**, additive manufacturing system **10** includes a computing device **12**, a powder delivery device **14**, an energy delivery device **16**, a powder flow monitoring system (PFMS) **18**, a melt pool monitoring system (MPMS) **56**, an optical system **54**, a stage **20**, a powder source **42**, a powder source mass sensor **44**, a topology sensor **48**, and in some examples, a machining device **52**.

[0032] Stage **20** is configured to position component **22** during an additive manufacturing process. In some examples, stage **20** is movable relative to energy delivery device **16** and/or energy delivery device **16** is movable relative to stage **20**. Similarly, stage **20** may be movable relative to powder delivery device **14** and/or powder delivery device **14** may be movable relative to stage **20**. Stage **20** may be configured to selectively position and restrain component **22** in place relative to stage **20** during manufacturing of component **22**.

[0033] Powder source **42** is the source of powder for powder stream **30**. Powder source **42** may include any suitable container or enclosure, such as a hopper, configured to hold powder. Powder source **42** also may include mechanism for entraining the powder in a gas flow. For instance, powder source **42** may be coupled to a gas source, which provides a gas flowing through powder source **42** and entraining powder within the gas flow. Additionally, or alternatively, powder source **42** may include an agitator configured to agitate the powder and increase entrainment of the powder in the gas stream. System **10** may include a powder source mass sensor **44** associated with powder source **42**. Powder source mass sensor **44** may be configured to quantify loss of mass in the powder source **42** or, alternatively, a mass flow out of powder source **42**.

[0034] Powder source **42** is fluidically coupled to powder delivery device **14** via a flow path **46**. Flow path **46** may include any suitable structure(s) defining an enclosed flow between powder source **42** and powder delivery device, including conduit, pipe, tubes, or the like. Although not shown in FIG. **1**, for at least part of flow path **46** between powder source **42** and nozzles of powder delivery device **14**, flow path **46** may split into multiple, parallel sections, e.g., one for each nozzle. Further, although not shown in FIG. **1**, in some examples, flow path **46** may include one or more nozzles for controlling flow through flow path **46** as a whole or portions of flow path **46** (e.g., a section associated with a particular nozzle of powder delivery device **14**).

[0035] Powder delivery device **14** may be configured to deliver powder to selected locations of component **22** being formed via a powder stream **30**. Powder delivery device **14** may include one or more nozzles that each output powder, such that the combined powder defines powder stream **30** focused at a focus plane. As powder delivery device **14** is movable in the z-axis shown in FIG. **1** relative to component **22**, the focal plane of powder delivery device **14** also may be movable in the z-axis relative to component **22**, such that the focus plane may be controlled to be substantially coincident with build surface **28**.

[0036] In some examples, powder delivery device **14** may be mechanically coupled or attached to energy delivery device **16** to facilitate delivery of powder stream **30** and energy **34** for forming melt pool **32** to substantially the same location adjacent to component **22**. Energy delivery device

**16** may include an energy source, such as a laser source, an electron beam source, plasma source, or another source of energy that may be absorbed by component **22** to form a melt pool **32** and/or be absorbed by powder in powder stream **30** to be added to component **22**. Example laser sources include a CO laser, a CO.sub.2 laser, a Nd:YAG laser, or the like. In some examples, the energy source may be selected to provide energy with a predetermined wavelength or wavelength spectrum that may be absorbed by component **22** and/or the powder to be added to component **22** during the additive manufacturing technique.

[0037] In some examples, energy delivery device **16** also includes an energy delivery head, which is operatively connected to the energy source. The energy delivery head may aim, focus, or direct energy **34** toward predetermined positions at or adjacent to a surface of component **22** during the additive manufacturing technique. As described above, in some examples, the energy delivery head may be movable in at least one dimension (e.g., translatable and/or rotatable) under control of computing device **12** to direct the energy toward a selected location at or adjacent to a surface of component **22**.

[0038] As shown in FIG. **1**, energy delivery device **16** may be arranged or configured such that energy **34** and powder stream **30** both exit from a common deposition head and are directed toward build surface **28**. For instance, energy **34** may pass through a central channel within the deposition head and exit a central aperture in the deposition head, while fluidized powder may flow through internal channels and powder nozzle(s) for forming powder stream **30** and directing powder stream **30** toward build surface **28**. At least some of the powder in powder stream **30** may impact a melt pool **32** in component **22**, and at least some of the powder that impacts melt pool **32** may be joined to component **22**.

[0039] During additive manufacturing, component **22** is built up by adding material to component **22** in sequential layers. The final component is composed of a plurality of layers of material. Energy delivery device **16** may direct energy **34** at first layer **24** to form melt pool **32**. Powder delivery device **14** may deliver powder stream **30** to melt pool **32**, where at least some of the powder at least partially melts and is joined to first layer **24**. Melt pool **32** cools as energy **34** is no longer delivered to that location of first layer **24** (e.g., due to energy delivery device **16** scanning energy **34** over the surface of first layer **24**). The temperature and cooling rate of melt pool **32** and the surrounding areas of first layer **24** affect the microstructure of the component **22** formed using the additive manufacturing technique.

[0040] System **10** may include both mass flow monitoring and heat flow monitoring. To provide mass flow monitoring, system **10** may include powder flow monitoring system (PFMS) **18**. PFMS **18** is configured to image at least a portion of powder stream **30** to detect powder flowing between powder delivery device **14** and build surface **28**. For example, PFMS **18** may include an illumination device and an imaging device. In some examples, the illumination device may include one or more light sources. For instance, the illumination device may include one or more structured light devices, such as one or more lasers. The illumination device is configured to illuminate a plane of powder stream **30** at image plane **38**, e.g., a plane substantially perpendicular to an axis extending between powder delivery device **14** and build surface **28**. The imaging device of PFMS **18** is configured to image at least some of the illuminated powder. The imaging device may have a relatively high data acquisition speed (e.g., frame rate), such greater than 1000 Hz. To provide heat flow monitoring, system **10** may include melt pool monitoring system (MPMS) **56**. MPMS **56** is configured to image at least a portion of melt pool **32** to detect parameters, such as size, temperature, or shape, of melt pool **32**. For example, MPMS may be communicatively coupled to optical system **54** for observing thermal emissions around melt pool **32** and a thermal camera for monitoring a size and/or temperature of melt pool **32**.

[0041] Optical system **54** may include an imaging device and an associated optical train, which senses emissions at or near component **22** during the additive manufacturing technique. For example, optical system **54** may include a visible light imaging device, an infrared imaging device, or an imaging device that is configured (e.g., using a filter) to image a specific wavelength or wavelength range. The optical train may include one or more reflective, refractive, diffractive optical components configured to direct light to the imaging device. For example, the optical train may be configured to direct light from near component **22** and/or melt pool **32** to the imaging device.

[0042] System **10** further includes a topology sensor **48**. Topology sensor **48** is configured to generate topological data representative of a topology of build surface **28**. Topological data may include any absolute or relative quantitative dimensional measurement of features on build surface **28**. As will be discussed further below, in addition to representing a topology of build surface **28**, topological data may also be used to at least partially determine a morphology of a build layer that is indicated by the topology of the build layer, alone or in combination with other measured or known parameters. As a result, topological data obtained for intermediate layers of component **22** may be used to select machining parameters for machining portions of the intermediate layers.

[0043] Topology sensor **48** may include any of a variety of sensors that utilize optical profilometry techniques including, but not limited to, interference profilometry, focus detection profilometry, pattern projection profilometry, or the like. In some examples, topology sensor **48** includes a laser and a sensor (e.g., an imaging device), which senses laser light reflected by the structure being imaged (e.g., melt pool **32** and the added material). The laser may have a defined wavelength, which may affect the resolution of the topology sensor **48**. In some examples, the wavelength and sensor may be selected such that the resolution of topology sensor **48** is as great as about 10 microns (e.g., about 6 microns). In some examples, topology sensor **48** may be positioned substantially directly above component **22** and may include an interferometer, which provides depth information based on the time from outputting a laser pulse to the sensing of the reflected light. In other examples, topology sensor **48** may be positioned at an offset with respect to component **22** such that the sensor senses depth information without using an interferometer.

[0044] Additive manufacturing system **10** may include machining device **52**. Machining device **52** may be configured to adaptively machine component **22** during a concurrent or subsequent subtractive manufacturing technique as part of the additive manufacturing process. In some examples, machining device **52** is configured to machine component **22** during initial fabrication of component **22**. For example, machining device **52** may be configured to selectively machine portions of a particular layer **24**, **26**, after deposition of the respective layer. In some examples, machining device **52** is configured to machine component after initial fabrication of component **22**. For example, machining device **52** may be configured to selective machine portions of a surface and/or subsurface of component **22** after deposition of layers of component **22** is complete. A variety of machining devices may be used including, but not limited to, cutting machines, milling machines, grinding machines, turning machines (e.g., lathes), drilling machines, boring machines, or the like.

[0045] Computing device **12** is configured to control components and operation of system **10** and may include, for example, a desktop computer, a laptop computer, a workstation, a server, a mainframe, a cloud computing system, or the like. Computing device **12** may be communicatively coupled to, and configured to control, powder delivery device **14**, energy delivery device **16**, PFMS **18**, optical system **54**, MPMS **56**, stage **20**, powder source **42**, powder source mass sensor **44**, and/or topology sensor **48** using respective communication connections. Although FIG. **1** illustrates a single computing device **12** and attributes all control and processing functions to that single computing device **12**, in other examples, system **10** may include multiple computing devices **12**, e.g., a plurality of computing devices **12**.

[0046] Computing device **12** may be configured to control operation of powder delivery device **14**, energy delivery device **16**, adjustable z-stage **40**, stage **20**, and/or topology sensor **48** to position component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, MPMS **56**, and/or topology sensor **48** during additive manufacturing processes. For example, as described above, computing device **12** may control stage **20** and powder delivery device **14**, energy delivery device **16**, adjustable z-stage **40** and/or topology sensor to translate and/or rotate along at least one axis to position component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, MPMS **56**, and/or topology sensor **48**. Positioning component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, MPMS **56**, and/or topology sensor **48** may include positioning a predetermined surface (e.g., a surface to which material is to be added) of component **22** in a predetermined orientation relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, MPMS **56**, and/or topology sensor **48**.

[0047] Computing device **12** may be configured to control system **10** to deposit layers **24** and **26** to form component **22** based on a set of deposition parameters. The set of deposition parameters may include energy, feed, and motion parameters that are configured to produce layers **24**, **26**, having various physical parameters, such as a height of layers **24**, **26**, and a density of layers **24**, **26**. For example, the set of deposition parameters may include power, beam diameter, beam profile, and wavelength of energy delivery device **16**; powder feed rate and gas feed rate of powder delivery device **14**; scan speed and deposition path of stage **20** relative to energy delivery device **16** and powder delivery device **14**; or any other operating parameters that may affect an amount and/or quality of material formed as layers **24**, **26**.

[0048] Computing device **12** may be configured to select deposition parameters that are configured to generate layers **24**, **26**, according to the desired physical parameters. As shown in FIG. **1**, component **22** may include a first layer **24** and a second layer **26**, although many components may be formed of additional layers, such as tens of layers, hundreds of layers, thousands of layers, or the like. Component **22** in FIG. **1** is simplified in geometry and the number of layers compared to many components formed using additive manufacturing techniques. Although techniques are described herein with respect to component **22** including first layer **24** and second layer **26**, the technique may be extended to components **22** with more complex geometry and any number of layers.

[0049] To form component **22**, computing device **12** may control powder delivery device **14** and energy delivery device **16** according to the set of operating parameters to form, on a surface **28** of first layer **24** of material, a second layer **26** of material. Computing device **12** may control energy delivery device **16** to deliver energy **34** to a volume at or near surface **28** to form melt pool **32**. For example, computing device **12** may control the relative position of energy delivery device **16** and stage **20** to direct energy to the volume. Computing device **12** also may control powder delivery device **14** to deliver powder stream **30** to melt pool **32**. For example, computing device **12** may control the relative position of powder delivery device **14** and stage **20** to direct powder stream **30** at or on to melt pool **32**.

[0050] Computing device **12** may control powder delivery device **14** and energy delivery device **16** to move energy **34** and powder stream **30** along build surface **28** in a pattern until layer **26** is complete. Computing device **12** may then control a z-axis position of stage **20** and/or powder delivery device **14** and energy delivery device **16** such that melt pool **32** will be formed on surface **36** of second layer **26**, and may control powder delivery device **14** and energy delivery device **16** to move energy **34** and powder stream **30** along build surface **28** in a pattern until layer **26** is complete.

[0051] System **10** may be configured with various in-situ monitoring techniques, including mass flux monitoring, heat flux monitoring, and topology monitoring, to control an additive manufacturing process and a subsequent subtractive manufacturing process. FIG. **2** is a process flow diagram illustrating a mass flux and heat flux monitoring and control technique. The technique of FIG. **2** may be implemented by system **10** of FIG. **1** and will be described with concurrent reference to FIG. **1**. However, it will be appreciated that system **10** may perform other techniques and the technique of FIG. **3** may be performed by other systems.

[0052] Computing device **12** may be configured to control a powder feed rate output by powder source **42** (see top left of FIG. **2**). For instance, computing device **12** may be configured to control an agitator of powder source **42**, a gas flow rate of gas flowing through powder source **42**, a position of one or more valves within flow path **46**, or the like to control a powder feed rate output by powder source **42**.

[0053] Computing device **12** may be configured to receive data from one or more mass flow monitoring sensors, including PFMS **18** and/or powder source mass sensor **44**. Data received from powder source mass sensor **44** indicates a mass flow of powder from powder source **42** to powder delivery device. Data from PFMS **18** indicates a mass flow of powder in powder stream **30** between powder delivery device **14** to adjacent melt pool **32**. In some examples computing device **12** may be configured to further receive topological data from topology sensor **48**. Data from topology sensor **48** indicates a topology of build surface **28**, and may further indicate powder mass captured by melt pool **32** and added to component **22**.

[0054] Computing device **12** may calculate one or more mass flow-related metrics based on the data received from PFMS **18**, powder source mass sensor **44**, and/or topology sensor **48**. For example, one or more computing devices **12** may determine a capture efficiency by determining a fraction or percentage of powder from powder stream **30** that is captured by melt pool **32** and added to component **22**, e.g., by dividing the powder mass captured by melt pool **32**, as determined based on data from topology sensor, into the mass flow determined based on data received from PFMS **18**. Further, computing device **12** may determine an overall mass flux using the data received from PFMS **18**, powder source mass sensor **44**, and/or topology sensor **48**. One or more computing devices **12** then may use the overall mass flux as an input to the control algorithm used to control the powder feed rate output by powder source **42** (see top left of FIG. **2**).

[0055] Similarly, computing device may be configured to control energy delivery device **16** to deliver energy **34** to first layer **24** to establish a given heat input (see bottom left of FIG. **2**). For example, one or more computing device **12** may control one or more operating parameters of energy delivery device **16**, such as intensity, pulse rate, pulse width, or the like; one or more positional parameters related to energy delivery device **16**, such as dwell time at a location, a movement rate relative to first layer **24**, an overlap between adjacent passes of energy **34** across first layer **24**, a pause time between adjacent passes of energy **34** across first layer **24**, or the like to control heat input to system **10** (e.g., to melt pool **32** and component **22**).

[0056] Computing device **12** may be configured to receive data from one or more heat sensors, such as optical system **54** and/or MPMS **56**. Computing device **12** may determine a cooling rate and associated heat from using data from optical system **54** and may determine a heat input into component using a size and/or temperature of melt pool **32** as observed by melt pool monitor. Computing device **12** may be configured to determine an overall heat flux using these data. Computing device **12** may then use the overall heat flux as an input to the control algorithm used to control the energy delivery by energy delivery device **16** (see top left of FIG. **2**). In some examples, computing device **12** may also use the deposit topology (captured powder mass) and/or capture efficiency metric in the determination of the heat flux, as the added powder mass and quench effects associated with the captured powder affect the cooling rate.

[0057] During an additive manufacturing process, various defects or deviations may occur in build surface **28**. These defects or deviations may occur due to changes in operating and material parameters including, but not limited to: powder type, shape, size, distribution, and porosity; laser power; powder flow rate; scan speed, pattern, and length; gas flow rate and direction; gas flow composition; scan pattern; and any other parameter in which deviation may introduce anomalies in formation of a layer. Defects in layers **24**, **26**, may at least partly manifest as topological deviations in build surface, such that a topology of build surface **28** may indicate the presence of the defects. Defects or deviations may manifest as local defects that only affect a particular portion of a layer, such as due to a temporary change in deposition conditions within a particular layer, or may manifest as global defects that affect an entire layer, such as a gradual change in deposition conditions over a sequence of layers.

[0058] In some instances, defects may include surface defects. Surface defects may include defects that are present at and limited to build surface **28**. As one example, a surface defect may include a planar deviation from a plane of build surface **28**, such as variation in thickness of the layer. Such planar deviation may occur, for example, due to variations in powder flow rate or warping due to spatial or temporal variation in temperature. As another example, a surface defect may include surface roughness in build surface **28**, such as localized projections in build surface **28**. Such surface roughness may occur, for example, due to variations in powder size or spatter.

[0059] In some instances, defects may include sub-surface defects. Sub-surface defects may include defects that are present at build surface **28**, but may extend into the layer. As one example, a sub-surface defect may include pores in the layer that manifest as cavities in build surface **28**. Such pores may occur, for example, due to lack of fusion of particles or gas trapped during the deposition. As another example, a sub-surface defect may include inclusions, such as large or foreign powder, that manifest as irregularly-shaped embedded particles. Such inclusions may occur, for example, due to contamination or inconsistent size of powder. As another example, a sub-surface defect may include unfused particles that may manifest as shallow trenches. Such unfused particles may occur, for example, due to excessive powder flow rate or inadequate laser power.

[0060] In some instances, defects may include thickness deviations. Thickness deviations may include deviations that are present across build surface **28**. As one example, a thickness deviation may include a gradual change in thickness between layers that manifests as a thicker or thinner layer. Such gradual change in thickness may occur, for example, due to a change in temperature of the intermediate build, resulting in gradually increasing thicknesses of each subsequent layer, or a change in build strategy of a particular layer due to a geometry of component **22**, resulting in different deposition conditions for a same set of deposition parameters as the underlying layers.

[0061] Computing device **12** may be configured to receive topological data from topology sensor **48** that includes an indication of these defects or deviations. For example, topology sensor **48** may continuously scan build surface **28** during deposition of a respective layer **24**, **26**, or may scan build surface **28** after deposition of the respective layer **24**, **26**, is complete. In the example of FIG. **1**, prior to formation of second layer **26** of material on first layer **24** of material, computing device **12** may receive topological data for first layer **24** of material from topology sensor **48**. This topological data may represent a topology of build surface **28**. The topology of build surface **28** may include local features or parameters, such as micron-scale features or roughness, or global features or parameters, such as planarity. The topological data may include a topology for each layer of the plurality of layers, and may be received in real-time as build surface **28** is scanned, once component **22** is fabricated, or at any other time prior to machining.

[0062] In some examples, additive manufacturing system **10** may be configured to adaptively machine component **22** based on topological data captured during initial fabrication of component **22**. Computing device **12** may be configured to identify differences between the topological data and specification data representative of a set of tolerances of build surface **28**. Differences between the as-deposited topology of build surface **28** represented by the topological data and the desired topology of build surface **28** may indicate a defect, deviation, or other anomaly. In some examples, computing device **12** may identify a location of the identified difference in the topological data. For example, computing device **12** may form a model of the as-deposited component **22** in which the location is marked. In some examples, computing device **12** may further characterize the identified difference based on various spatial characteristics of the identified difference, such as a pattern, shape, or magnitude of the identified difference or differences. Computing device **12** may control machining device **52** to machine build surface **28** and/or component **22** based on the identified differences.

[0063] In some examples, computing device **12** may be configured to control machining device **52** to machine component **22** after initial fabrication of component **22**. As explained above, portions of component **22** may require further machining to achieve desired dimensions or finish, and portions of component **22** that include the defects may be more susceptible to damage than portions of component **22** that are deposited within specification. Computing device **12** may determine a toolpath for machining component **22** and operate machining device **52** along the toolpath such that a portion of component **22** that includes a defect may be machined differently than portions that do not include the defect. For example, computing device **12** may identify a sub-surface defect that is undetectable by a final optical scan of component **22**, but that is indicated as including an identified difference by the topological data. Computing device **12** may control machining device **52** according to more sensitive conditions (e.g., slower) such that the sub-surface defect may be less likely to undergo damage in response to the machining operation.

[0064] FIG. **3**A is a flowchart illustrating an example method for fabricating component **22** using topological data, including machining component **22** after initial fabrication is complete. The example method of FIG. **3**A includes controlling energy delivery device **16** and powder delivery device **14** (**60**) to deposit a layer on build surface **28** based on a set of deposition parameters (**62**). For example, computing device **12** may send control signals to each of energy deliver device **16** and powder delivery device **14**. In response to receiving the control signals, energy delivery device **16** delivers energy to build surface **28** of component **22** to form melt pool **32** in build surface **28** (**64**), and powder delivery device **14** directs powder stream **30** toward melt pool **32** (**66**).

[0065] The example method of FIG. **3**A includes generating, by topology sensor **48**, topological data representative of build surface **28** (**68**). For example, topology sensor **48** may use one or more optical techniques to scan build surface **28** and capture topological data representing a topology of build surface **28**. The topological data may include a topology for each layer of the plurality of layers. The example method of FIG. **3**A includes receiving topological data from topology sensor **48** of additive manufacturing system **10** (**70**). For example, computing device **12** may receive the topological data in real-time as build surface **28** is scanned, once component **22** is fabricated, or at any other time prior to machining.

[0066] The example method of FIG. **3**A includes identifying, by computing device **12**, differences between the topological data and specification data representative of a set of tolerances of build surface **28** (**72**). The identified differences may represent defects or other anomalies that may affect how the subsequent machining operation is performed, but that would otherwise go unnoticed if not for the captured topological data of each layer **24**, **26**.

[0067] The example method of FIG. **3**A includes controlling machining device **52** to machine one or more portions of component **22** based on the identified differences (**74**). For example, while component **22** may be initially fabricated to dimensions similar to nominal dimensions of a finished part, such dimensions may require additional machining, particularly for portions of component **22** that may be configured with particular surface properties or that may be subject to small clearances or tolerances. As a result, computing device **12** may control machining device **52** based on a set of machining parameters that account for these identified differences to machine portions of component **22** that may be out of specification until component **22** is within tolerance. In response to receiving the control signals, machining device **52** machines component **22** based on the identified differences (**76**).

[0068] FIG. **3**B is a flowchart illustrating an example method for generating machining data using topological data. The example method of FIG. **3**B may be performed by computing device **12**, such that computing device **12** may be configured to perform any of the steps of FIG. **3**B. The example method includes receiving topographical data for build surface **28** (**70**), such as described in step **70** of FIG. **3**A above. The example method of FIG. **3**B further includes receiving specification data for build surface **28** (**80**). The specification data may represent an anticipated or desired topology of build surface **28** based on the set of deposition parameters selected for the particular layer. For example, the specification data may include a desired build height of the particular layer, such as relative to an underlying layer or relative to the build height at that particular layer.

[0069] FIG. **3**C is a cross-sectional side view diagram illustrating an intermediate build of a component that includes topological defects **92**A and **92**B (referred to collectively as "defects **92**"), such as may be indicated by the topological data. Defects **92** may include any defects in or on layer **90**B that can be indicated by a topology of a build surface **94** of layer **90**B. In the example of FIG. **3**C, defects **92** are each indicated by a projection; however, other topological variations may indicate defects **92**, such as depressions, cavities, ridges, troughs, or any other difference in topology detectable using optical detection techniques.

[0070] Referring back to FIG. **3**B, the example method includes identifying differences between the topological data and specification data representative of a set of tolerances of build surface **28** (**72**), such as described in step **72** of FIG. **3**A above. In some examples, the identified differences include a spatial location of a topological variation that exceeds a tolerance, such as x-, y-, and z-dimensions of the topological variation at x- and y-coordinates on or in the particular layer. In some instances, the topological variation may include only the portion of the layer that exceeds the tolerance. For example, a defect that results from highly localized over-deposition, such as surface roughness, may be limited to that portion of the layer outside the tolerance. In some instances, the topological variation may include an entire portion of the layer underlying the portion of the layer that exceeds the tolerance. For example, a defect that results from unfused powder may include the entire layer at that portion, rather than just the surface portion. FIG. **3**D is a cross-sectional side view diagram illustrating identification of topological defects in an intermediate build of a component. In the example of FIG. **3**D, defects **92**A and **92**B are indicated as respective regions **93**A and **93**B of identified differences.

[0071] In some examples, the identified differences may be further categorized based on various characteristics of the identified differences. For example, computing device **12** may analyze characteristics of each region **93** such as a magnitude (e.g., height or depth) of a corresponding defect **92**, a size (e.g., width or length) of a corresponding defect, a pattern (e.g., relative distance) of two or more defects **92**, a shape of a corresponding defect **92**, or any other characteristic of defect **92** that may differentiate a type of defect. The type of defect may correspond to a material property that results from the defect. For example, a warping defect may indicate that the portion of build surface **28** has undergone a different thermal treatment, while a porous defect may indicate that the portion has a lower (e.g., different) density. Types of defects that may be categorized may include, but are not limited to, porosity, unfused powder, cavities, warping, and the like.

[0072] Referring back to FIG. **3**B, the example method includes generating model data that includes the identified differences (**82**). The model data may represent an as-fabricated model of component **22**, including an internal topology and/or morphology of build surfaces **28** that is based on the topological data and that includes the identified differences. For example, the identified differences may be indicated as spatial locations within the model data. FIG. **3**E is a cross-sectional side view diagram illustrating generation of machining data that accounts for topological defects. Regions **93**A and **93**B are indicated within the model data, such that the presence and/or material characteristics of regions **93**A and **93**B can be considered when generating the machining data.

[0073] Referring back to FIG. **3**B, the example method includes generating machining data based on the model data (**84**). Machining data may include a toolpath that corresponds to one or more excess portions of component **22** to be removed through machining, as well as the sets of machining conditions at those excess portions. Computing device **12** may compare the model data with the final part data to identify an excess portion of component **22** to be removed. In the example of FIG. **3**E, such excess portion is indicated by machining line **87**. This excess portion includes a portion of region **93**A that is beneath a surface of component **22**, and that may not otherwise be indicated by topological data of the surface of component **22**.

[0074] Computing device **12** may further select the set of machining parameters for the excess portion. In some examples, the presence of defects **92** may indicate a general set of machining conditions. For example, regardless of a type of defect, presence of a defect within the excess portion may indicate a set of machining parameters that are less likely to cause damage or exacerbate existing damage caused by the defect. In the example of FIG. **3**E, computing device **12** may select a first set of machining parameters **98**A for portions of component **22** outside of regions **93**A and a second set of machining parameters **98**B for portions of component **22** within region **93**A. The second set of machining parameters **98**B may include a slower speed of an abrasive or cutting surface, a finer feature size of an abrasive surface, a shallower angle of an abrasive or cutting surface, selection of a newer cutting tool, a modified toolpath, or any other machining condition that may be less likely to damage a portion of component **22** having a lower material quality (e.g., higher porosity, lower strength, higher brittleness, etc.).

[0075] In some examples, the type of defects **92** may indicate a particular set of machining conditions. In the example of FIG. **3**E, computing device **12** may select first set of machining conditions **98**A for portions of component **22** outside of regions **93**A and second set of machining conditions **98**B for portions of component **22** within region **93**A that are particular to the type of defect identified for defect **92**A. A type of defect within the excess portion may indicate the use of a particular set of machining parameters that are less likely to cause damage to that particular type of defect. As one example, an unfused powder defect may be more brittle than portions that include fused powder, such that a lower speed of an abrasive surface may be used. As another example, a warping defect may be harder than portions that were not subject to the same high local temperature, such that a higher speed of an abrasive surface and/or a higher angle of an abrasive or cutting surface may be used.

[0076] Referring back to FIG. **3**B, the example method includes outputting machining data (**86**). In response to receiving the machining data, machining device **52** may machine portions of component **22** that do not include defects according the first set of machining conditions and machine portions of component **22** that include region **93**A according to the second set of machining parameters. As a result, component **22** may be machined more efficiently and/or effectively compared to machining processes that do not utilize topological data to determine subsequent machining conditions.

[0077] In some examples, computing device **12** may be configured to control machining device **52** to machine layers **24**, **26** during initial fabrication of component **22**. For example, portions of component **22** that include a defect may have inferior material properties that may propagate to subsequent layers. Computing device **12** may evaluate whether the identified difference exceeds a threshold, such as for machinability or severity. If the identified difference exceeds the threshold, computing device **12** may control machining device **52** to machine the identified difference to within tolerance, such that component **22** may have fewer or less severe internal defects.

[0078] FIG. **4**A is a flowchart illustrating an example method for fabricating component **22** using topological data, including machining layers of component **22** during initial fabrication of component **22**. The example method of FIG. **4**A includes controlling energy delivery device **16** and powder delivery device **14** (**100**) to deposit a layer on build surface **28** based on a set of deposition parameters (**102**). For example, computing device **12** may send control signals to each of energy deliver device **16** and powder delivery device **14**. In response to receiving the control signals, energy delivery device **16** delivers energy to build surface **28** of component **22** to form melt pool **32** in build surface **28** (**104**), and powder delivery device **14** directs powder stream **30** toward melt pool **32** (**106**).

[0079] The example method of FIG. **4**A include generating, by topology sensor **48**, topological data representative of build surface **28** (**108**). For example, topology sensor **48** may use one or more optical techniques to scan build surface **28** and capture topological data. The topological data may include a topology for a particular layer of the plurality of layers that has just been deposited. The example method of FIG. **3**A includes receiving topological data from topology sensor **48** of additive manufacturing system **10** (**100**). For example, computing device **12** may receive the topological data in real-time as build surface **28** is scanned or at any other time prior to deposition of a subsequent layer on build surface **28**.

[0080] The example method of FIG. **4**A includes identifying, by computing device **12**, differences between the topological data and specification data representative of a set of tolerances of build surface **28** (**112**). The identified differences may represent defects or other anomalies that may affect how the subsequent machining operation is performed, but that would otherwise go unnoticed if not for the captured topological data of each build surface **28**.

[0081] The example method of FIG. **4**A includes controlling, by computing device **12**, machining device **52** to machine one or more portions of build surface **28** based on the identified differences (**114**). For example, computing device **12** may control machining device **52** to machine portions of build surface **28** that may be out of specification until build surface **28** is within tolerance. In response to receiving the control signals, machining device **52** may machine build surface **28** based on the identified differences (**116**). After machining is complete, the example method includes depositing additional layers on build surface **28** until initial fabrication of component **22** is complete.

[0082] FIG. **4**B is a flowchart illustrating an example method for generating machining data using topological data. The example method of FIG. **3**B may be performed by computing device **12**, such that computing device **12** may be configured to perform any of the steps of FIG. **4**B. The example method includes receiving topographical data for build surface **28** (**110**), such as described in step **110** of FIG. **4**A above. The example method of FIG. **4**B further includes receiving specification data for build surface **28** (**120**). The specification data may represent an anticipated or desired topology of build surface **28** based on the set of deposition parameters selected for the particular layer.

[0083] FIG. **4**C is a cross-sectional side view diagram illustrating an intermediate build of a component that includes topological defects **132**A and **132**B (referred to collectively as “defects **132**”), such as may be indicated by topological data. Defects **132** may include any defects on layer **90**B that can be indicated by a topology of a build surface **134**A of layer **130**B and removed through a machining process. In the example of FIG. **4**C, defects **132** are indicated by a projection; however, other topological variations may indicate defects **132**, such as ridges, overhangs, or any other difference in topology detectable using optical detection techniques and actionable using subtractive manufacturing techniques.

[0084] Referring back to FIG. **4**B, the example method includes identifying differences between the topological data and specification data representative of a set of tolerances of build surface **28** (**112**), such as described in step **112** of FIG. **4**A above. In some examples, the identified differences include a spatial location of a topological variation that exceeds a tolerance. In some instances, the spatial location may include only the portion of the layer that exceeds the tolerance. For example, a defect that results from localized over-deposition may be limited to that portion of the layer outside the tolerance.

[0085] The example method of FIG. **4**B includes evaluating whether each of the identified differences exceeds a machinability threshold (**122**). The machinability threshold may represent a threshold for which removal of the identified difference is not sufficiently valuable and/or feasible based on a capability of machining device **52**. For example, an identified difference may not be of a sufficient magnitude (e.g., height or size) to adversely affect an integrity of component **22**. Additionally or alternatively, removal of the identified difference may not be within a machining tolerance of machining device **52** and/or sufficiently valuable for the amount of time spent removing the material or potential risk of causing damage to build surface **28**.

[0086] The example method of FIG. **4**B includes generating model data that includes the identified differences (**124**). The model data may represent an as-deposited model of build surface **28** that is based on the topological data and that includes the identified differences with respect to the machinability threshold. For example, the identified differences may be indicated as spatial locations within the model data. FIG. **4**D is a cross-sectional side view diagram illustrating identification of topological defects in an intermediate build of a component. In the example of FIG. **4**D, only defect **132**A is indicated as a respective regions **133**A that includes an identified difference that exceeds a machining threshold **135**.

[0087] Referring back to FIG. **4**B, the example method includes generating machining data based on the model data (**126**). Machining data may include a toolpath that corresponds to regions **133** to be removed through machining. FIG. **4**E is a cross-sectional side view diagram illustrating machining of topological defects **132**. Defect **132**A is machined to form a new build surface **134**B within a desired tolerance. In the example of FIG. **4**E, defect **132**A is illustrated as being machined to at or below machinability threshold **135**; however, in other examples, defect **132**A may be machined to a different threshold. For example, defect **132**A may be fully machined such as to be in a same plane as a remaining build surface **134**B outside of defects **132**A and **132**B.

[0088] Once the identified differences are machined, the example method may include further depositing any subsequent layers on build surface **28**. FIG. **4**F is a cross-sectional side view diagram illustrating deposition of an overlying layer **130**C on machined layer **130**B. As a result, defects in an underlying layer **130**B may not propagate, or may propagate to a lesser extent, in any subsequent layers **130**C.

[0089] In some examples, additive manufacturing system **10** may be configured to adaptively deposit subsequent layers of component **22** based on topological data captured during initial fabrication of component **22**. A particular set of deposition parameters may result in a layer having a different height than predicted based on the set of deposition parameters. For example, as layers are deposited, small differences between the predicted and deposited height of the layers may propagate over multiple layers, such that final component **22** requires substantial material removal to bring component **22** into specification. These small differences in height may be due to deposition parameters that, individually or collectively, may vary over the course of a build and/or over the course of operation of system **10**.

[0090] Computing device **12** may be configured to determine whether a layer or sequence of layers exceed the threshold using the topological data and adjust deposition for subsequent layers that account for the set of parameters, such that component **22** more closely matches a target set of dimensions, thereby reducing or eliminating subsequent machining. For example, computing device **12** may be configured to use machine learning techniques that evaluate a set of deposition parameters with respect to the topological data to adjust the set of deposition parameters for subsequent layers.

[0091] FIG. **5**A is a flowchart illustrating an example method for fabricating component **22** using topological data. The example method of FIG. **5**A includes controlling energy delivery device **16** and powder delivery device **14** (**140**) to deposit a layer on build surface **28** based on an initial set of deposition parameters (**142**). For example, computing device **12** may send control signals to each of energy deliver device **16** and powder delivery device **14**. In response to receiving the control signals, energy delivery device **16** delivers energy to build surface **28** of component **22** to form melt pool **32** in build surface **28** (**144**), and powder delivery device **14** directs powder stream **30** toward melt pool **32** (**146**).

[0092] The example method of FIG. **5**A include generating, by topology sensor **48**, topological data representative of build surface **28** (**148**). For example, topology sensor **48** may use one or more optical techniques to scan build surface **28** and capture topological data. The topological data may include a topology for a particular layer of the plurality of layers. The example method of FIG. **5**A includes receiving topological data from topology sensor **48** of additive manufacturing system **10** (**70**). For example, computing device **12** may receive the topological data in real-time as build surface **28** is scanned, prior to deposition of a subsequent layer, or at any other time prior to completion of component **22**.

[0093] The example method of FIG. **5**A includes identifying, by computing device **12**, differences between the topological data and specification data representative of a set of tolerances of build surface **28** (**152**). The identified differences may represent a change in thickness of the particular layer or a sequence of layers. For example, the identified differences may result from an anomalous change in thickness of a particular layer that exceeds a threshold for that particular layer, or may result from a gradual change in thickness over a sequence of layers that exceeds a threshold for the overall build at that particular layer.

[0094] The example method of FIG. **5**A includes determining an adjusted set of deposition parameters for a subsequent layer based on the identified differences (**154**). The adjusted set of deposition parameters may be selected to form a subsequent layer or sequence of layers that are within specification and/or that will bring the build into specification. For example, an overlying layer may be deposited that compensates for deviation of a particular layer, or a series of overlying layers may be deposited that compensate for the deviation of a previous sequence of layers. The example method of FIG. **5**A includes controlling energy delivery device **16** and powder delivery device **14** (**140**) to deposit a layer on build surface **28** based on an initial set of deposition parameters (**142**).

[0095] FIG. **5**B is a flowchart illustrating an example method for generating subsequent layers using topological data. The example method of FIG. **5**B may be performed by computing device **12**, such that computing device **12** may be configured to perform any of the steps of FIG. **5**B. The example method includes receiving topographical data for build surface **28** (**150**), such as described in step **150** of FIG. **5**A above. The example method of FIG. **5**B further includes receiving specification data for build surface **28** (**160**). The specification data may represent an anticipated or desired topology of build surface **28** based on the set of deposition parameters selected for the particular layer. The specification data may include a target build height that corresponds to the anticipated or desired topology of build surface **28**. In some instances, this specification data may represent a build height of the particular layer. For example, the specification data may represent an anticipated or desired difference in build level between the currently deposited layer and the previously deposited layer. In some instances, the specification data may represent a spatial location of build surface **28** for the particular layer in the build of component **22**. For example, the specification data may represent an anticipated or desired build height of the deposited layers. The specification data may also include one or more tolerances corresponding to a difference in the current build surface **28** to a build surface of a previous layer and/or a difference in the current build surface **28** to an anticipated build surface at the particular layer in the build. As will be described below, this tolerance may be predetermined, may be determined during the build, or both.

[0096] The example method of FIG. **5**B includes determining a build height of an as-deposited layer based on the topological data (**161**). For example, computing device **12** may determine a build height of the layer or a sequence of layers based on a relative difference in the topology of the layer and a reference, such as the topology of the underlying layer. In some examples, the build height may be an average build height of the layer. For example, computing device **12** may determine a spatial average of the topology of build surface **28**. In some examples, the build height may be a maximum and/or minimum build height of the layer.

[0097] FIG. **5**C is a cross-sectional side view diagram illustrating an intermediate build of component **22** that includes a topological deviation from a target build height of a build surface **174**A, such as may be indicated by the topological data. The intermediate build includes a first layer **170**A having a first build height **172**A and a second layer **170**B having a second build height **172**B, greater than the first thickness. As layers **170** are deposited, deposition conditions may change, resulting in layers having different thicknesses. Such topological deviation may occur despite input parameters remaining substantially the same. In some instances, such topological deviation may be due to an anomalous change in deposition conditions that only affects one to several layers. For example, a restart of the build between layers may result in a first layer after resumption of the build being different from an underlying layer. In some instances, such topological deviation may be due to a gradual change in deposition conditions over a sequence of layers. For example, a change in geometry of component **22** over the sequence of layers may cause different heating or cooling dynamics of the layers, resulting in layers having different thicknesses.

[0098] Referring back to FIG. **5**B, the example method includes identifying a difference between the build height of the layer and the target build height for the layer (**162**). The target build height for a layer may correspond to a target build surface relative to an underlying layer, such as determined from a desired thickness of the layer based on the specification data. Computing device **12** may compare the build height for the layer as indicated by the topological data with the target build height as indicated by or determined from the specification data to identify the difference. FIG. **5**D is a cross-sectional side view diagram illustrating identification of topological deviation in an intermediate build of component **22**. An excess portion **173** of layer **170**B may exceed a target build surface **175**A. Target build surface **175**A may represent the target build height of build surface **28**. In some examples, target build surface **175**A may be defined with respect to a previous deposition height of underlying layer **170**A, such that reference surface **175**A may represent a projected build surface for a target build height of layer **170**B. In some examples, reference surface **175**A may be defined with respect to an overall height of some or all underlying layers **170**, such that target build height **175**A may represent a target spatial position of build surface **174**A of layer **170**B.

[0099] Referring back to FIG. **5**B, the example method includes determining whether the identified difference exceed a threshold (**163**). As mentioned above, the specification data may include tolerances related to a thickness of a layer and/or a height of build surface **174**A for a particular layer within the build. This tolerance may account for instrument error of topology sensor **48** and/or random variation between layers that does not indicate a change in deposition conditions that requires compensation. Computing device **12** may determine whether excess portion **173** exceeds the tolerance to determine whether to modify the set of deposition parameters for any overlying layers.

[0100] In response to determining that the identified difference does not exceed the tolerance, the method includes outputting a current set of deposition parameters. For example, computing device **12** may continue to deposit subsequent layers until a layer has a build height that exceeds the tolerance. In response to receiving the initial set of deposition parameters, energy delivery device **16** and powder delivery device **14** may deposit the subsequent layer based on the initial set of deposition parameters.

[0101] In response to determining that the identified difference exceeds the tolerance, the example method includes determine an adjusted set of deposition parameters of a subsequent layer (**166**). The adjusted set of deposition parameters are different from the set of deposition parameters of an underlying layer. The adjusted set of deposition parameters may include at least one of a thickness of a respective layer, a power of the energy delivery device, a size of the melt pool, a feed rate of the powder stream, a travel speed of the powder stream relative to the build surface, or a tool path of melt pool along the build surface **28**. The method may include outputting the adjusted set of deposition parameters (**167**); in response, energy delivery device **16** and powder delivery device **14** deposit the subsequent layer based on the adjusted set of deposition parameters.

[0102] FIG. **5**E is a cross-sectional side view diagram illustrating deposition of a subsequent layer that accounts for the topological defects of the underlying layer. In some examples, the adjusted set of deposition parameters may be selected to compensate for a previous topological deviation in an underlying layer. For example, the adjusted set of deposition parameters for layer **170**C may result in layer **170**C having a lower thickness to compensate for a higher thickness of layer **170**B. Such compensation may be particularly useful for anomalous thickness deviations that may not manifest in subsequent layers. In some examples, the adjusted set of deposition parameters may be selected to compensate for a change in anticipated build height that results from a change in deposition conditions. For example, the adjusted set of deposition conditions for layer **170**C may result in layer **170**C having a build height that corresponds to a desired build height.

[0103] Referring back to FIG. **5**B, in some examples, determining an adjusted set of deposition conditions may include only determining an adjusted set of deposition conditions for a particular portion of a subsequent layer and using an existing set of deposition conditions for a remainder of the subsequent layer. For example, the topological data may indicate that only a portion of build surface deviates from the target build surface beyond the threshold. Computing device **12** may identify those portions and determine the adjusted set of deposition parameters for those portions. Computing device **12** may output the adjusted set of deposition parameters to cause energy delivery device **16** and powder delivery device **14** to deposit portions of the subsequent layer overlying the identified difference based on the adjusted set of deposition parameters and deposit portions of the subsequent layer that are not overlying an identified difference based on the initial set of deposition parameters.

[0104] In some examples, the adjusted set of deposition parameters may be selected using machine learning techniques. For example, a particular build height of a layer may be related to a combination of deposition parameters. The particular combination of deposition parameters may vary depending on a progression of a build, an environment in which the build is performed, a geometry of component **22**, or the like. Computing device may determine one or more relationships between a build height of the layer and one or more training sets of deposition parameters based on topological data. Computing device **12** may use the topological data as feedback for determining the build height produced by the training sets of deposition parameters. The topological data and training sets of deposition conditions may include topological data and sets of deposition conditions for other layers in the same build of component **22**, for other layers of other builds of similar components as component **22**, and/or for other layers of other builds for dissimilar components as component **22**. Computing device **12** may use machine learning algorithms to determine relationships between various sets of deposition parameters and determine the adjusted set of parameters for a particular layer based on these relationships.

[0105] In some examples, the method of FIG. **5**B includes determining whether the identified difference can be corrected through adjustment of deposition of subsequent layers (**165**). For example, the specification data may further include one or more thresholds related to a maximum permitted deviation of a build height from the target build height. This threshold may be related to an integrity of component **22**, as a substantial deviation may indicate a defect that may require correction. In response to determining that the identified difference cannot be corrected through deposition of subsequent layers, the method may include determining a set of machining parameters for machining at least a portion of build surface **28** that includes the respective identified difference (**168**) and outputting the set of machining parameters to a machining device (**169**). For example, the set of machining parameters may include a location of particular portions of build surface that exceed the threshold.

[0106] The evaluation described in steps **150** and **161-167** may continue for subsequent layers. For example, computing device **12** may receive topological data from topology sensor **48** for the subsequent layer deposited according to the previously adjusted set of deposition parameters (**150**). Computing device **12** may determine a build height of the subsequent layer based on the topological data for the subsequent layer (**161**), identify a difference between the build height of the subsequent layer and a target build height of the subsequent layer (**162**), determine whether the identified difference exceeds the tolerance (**163**) and control energy delivery device **16** and powder delivery device **14** to deposit the subsequent layer based on whether the identified difference exceeds the tolerance (**164**, **166**, **167**).

[0107] As explained above, additive manufacturing systems described herein may be configured to use machine learning processes for evaluating a set of deposition parameters with respect to topological data to adjust the set of deposition parameters for subsequent layers. FIG. **6**A is a conceptual diagram illustrating an example machine learning model according to one or more aspects of this disclosure. Machine learning model **200** may be an example of a machine learning model implemented by computing device **12**. Machine learning model **200** may be an example of a deep learning model, or deep learning algorithm, trained to determine deposition parameters for subsequent layers. Computing device **12** and/or another device, may train, store, and/or utilize machine learning model **200**, but other devices of system **10** may apply inputs to machine learning model **200**. In some examples, other types of machine learning and deep learning models or algorithms may be utilized. For example, a convolutional neural network model of ResNet-18 may be used. Some non-limiting examples of models that may be used for transfer learning include AlexNet, VGGNet, GoogleNet, ResNet50, or DenseNet, etc. Some non-limiting examples of machine learning techniques include Support Vector Machines, K-Nearest Neighbor algorithm, and Multi-layer Perceptron.

[0108] As shown in the example of FIG. **6**A, machine learning model **200** may include three types of layers. These three types of layers include input layer **202**, hidden layers **204**, and output layer **206**. Output layer **206** includes the output from the transfer function **205** of output layer **206**. Input layer **202** represents each of the input values X1 through X4 provided to machine learning model **200**. In some examples, the input values may include any of the values input into the machine learning model, as described above. For example, the input values may include topological data from topology sensor **48**, or other data received by computing device **12**, as described above. In addition, in some examples input values of machine learning model **200** may include additional data, such as other data that may be collected by or stored in system **200**.

[0109] Each of the input values for each node in the input layer **202** is provided to each node of a first layer of hidden layers **204**. In the example of FIG. **6**A, hidden layers **204** include two layers, one layer having four nodes and the other layer having three nodes, but fewer or greater number of nodes may be used in other examples. Each input from input layer **202** is multiplied by a weight and then summed at each node of hidden layers **204**. During training of machine learning model **200**, the weights for each input are adjusted to establish a relationship between topological data and deposition parameters that generate the layer represented by the topological data. In some examples, one hidden layer may be incorporated into machine learning model **200**, or three or more hidden layers may be incorporated into machine learning model **200**, where each layer includes the same or different number of nodes.

[0110] The result of each node within hidden layers **204** is applied to the transfer function of output layer **206**. The transfer function may be linear or non-linear, depending on the number of layers within machine learning model **200**. Example non-linear transfer functions may be a sigmoid function or a rectifier function. The output **207** of the transfer function may be a set of deposition parameters that correspond to the layer represented by the topological data and having a desired thickness.

[0111] As shown in the example above, by applying machine learning model **200** to input data such as image data, processing circuitry of computing device **12** is able to determine a relationship between deposition parameters that produce a layer based on the topological data. Computing device **12** may utilize such features to determine a local coordinate system, to map the local coordinate system to a global coordinate system, and/or to generate representations of the layer and/or layers.

[0112] FIG. **6**B is a conceptual diagram illustrating an example training process for a machine learning model according to one or more aspects of this disclosure. Process **210** may be used to train machine learning model **200**. Machine learning model **200** may be implemented using any number of models for supervised and/or reinforcement learning, such as but not limited to, an artificial neural network, a decision tree, naïve Bayes network, support vector machine, or k-nearest neighbor model, CNN, RNN, LSTM, ensemble network, to name only a few examples.

[0113] In some examples, computing device **12** or another device trains machine learning model **100** based on a corpus of training data **212**. Training data **212** may include, for example, previous topological data of a build over time, previous deposition parameter data, and/or the like. Previous deposition parameter data, for example, may include deposition parameters associated with previous additive manufacturing processes. In some examples, the topological data and/or deposition parameter data may be generated by executing a natural language processing application on content of a plurality of operational records to automatically extract relevant nor desired training data. For example, by using an NLP model, computing device **12** may capture potentially cofounding operation conditions or factors, which may be useful in determining a set of deposition parameters. Computing device **12** may use the NLP model to capture such details.

[0114] In some examples, training data **212** may include annotations identifying effects of deposition parameters indicated in topological data of training data **212**. Training data **212** may include data from past additive deposition processes performed on components having different geometries, formed from different materials, formed under different conditions, and/or the like.

[0115] While training machine learning model **200**, computing device **12** may compare **214** a prediction or classification with a target output **216**. Computing device **12** may utilize an error signal from the comparison to train (learning/training **218**) machine learning model **200**. Computing device **12** may generate machine learning model weights or other modifications which computing device **12** may use to modify machine learning model **200**. For example, computing device **12** may modify the weights of machine learning model **200** based on the learning/training **218**.

[0116] The techniques described in this disclosure may be implemented, at least in part, in hardware, software, firmware, or any combination thereof. For example, various aspects of the described techniques may be implemented within one or more processors, including one or more microprocessors, digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic circuitry, as well as any combinations of such components. The term “processor” or “processing circuitry” may generally refer to any of the foregoing logic circuitry, alone or in combination with other logic circuitry, or any other equivalent circuitry. A control unit including hardware may also perform one or more of the techniques of this disclosure.

[0117] Such hardware, software, and firmware may be implemented within the same device or within separate devices to support the various techniques described in this disclosure. In addition, any of the described units, modules or components may be implemented together or separately as discrete but interoperable logic devices. Depiction of different features as modules or units is intended to highlight different functional aspects and does not necessarily imply that such modules or units must be realized by separate hardware, firmware, or software components. Rather, functionality associated with one or more modules or units may be performed by separate hardware, firmware, or software components, or integrated within common or separate hardware, firmware, or software components.

[0118] The techniques described in this disclosure may also be embodied or encoded in an article of manufacture including a computer-readable storage medium encoded with instructions. Instructions embedded or encoded in an article of manufacture including a computer-readable storage medium encoded, may cause one or more programmable processors, or other processors, to implement one or more of the techniques described herein, such as when instructions included or encoded in the computer-readable storage medium are executed by the one or more processors. Computer readable storage media may include random access memory (RAM), read only memory (ROM), programmable read only memory (PROM), erasable programmable read only memory (EPROM), electronically erasable programmable read only memory (EEPROM), flash memory, a hard disk, a compact disc ROM (CD-ROM), a floppy disk, a cassette, magnetic media, optical media, or other computer readable media. In some examples, an article of manufacture may include one or more computer-readable storage media.

[0119] In some examples, a computer-readable storage medium may include a non-transitory medium. The term “non-transitory” may indicate that the storage medium is not embodied in a carrier wave or a propagated signal. In certain examples, a non-transitory storage medium may store data that can, over time, change (e.g., in RAM or cache).

[0120] Example 1: An additive manufacturing system includes an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool; a machining device configured to machine the build surface; one or more sensors includes receive the topological data from the at least one topology sensor for a plurality of layers; identify differences between the topological data and specification data representative of a set of tolerances of the build surface; control the energy delivery device and the powder delivery device to deposit the plurality of layers based on a set of deposition parameters; and control the machining device to machine the build surface based on the identified differences.

[0121] Example 2: The additive manufacturing system of example 1, wherein the computing device is configured to identify at least one of the identified differences as a defect corresponding to a different density.

[0122] Example 3: The additive manufacturing system of any of examples 1 and 2, wherein the plurality of layers includes a plurality of intermediate layers, and wherein at least one difference of the identified differences is identified in an intermediate layer of the plurality of intermediate layers.

[0123] Example 4: The additive manufacturing system of example 3, wherein the computing device is further configured to generate model data that includes the identified differences.

[0124] Example 5: The additive manufacturing system of example 4, wherein, to control the machining device, the computing device is configured to: generate machining data based on the model data; and output the machining data to the machining device.

[0125] Example 6: The additive manufacturing system of example 5, wherein the machining data includes: a toolpath; a first set of machining parameters for portions of the toolpath that do not include the identified differences; and a second set of machining parameters for portions of the toolpath that include the identified differences, wherein at least one machining parameter of the second set of machining parameters is different from the first set of machining parameters.

[0126] Example 7: The additive manufacturing system of example 6, wherein the at least one machining parameter includes a change in material removal rate.

[0127] Example 8: The additive manufacturing system of any of examples 3 through 7, wherein the computing device is configured to, for an intermediate layer of the plurality of intermediate layers: identify the at least one difference for the intermediate layer; control the machining device to machine the build surface based on the at least one difference; and after machining the build surface, control the energy delivery device and the powder delivery device to deposit a subsequent layer on the intermediate layer.

[0128] Example 9: The additive manufacturing system of any of examples 1 through 8,wherein the computing device is further configured to: evaluate whether at least one difference exceeds a machinability threshold; and in response to the at least one difference exceeding the machinability threshold, controlling the machining device to machine the build surface.

[0129] Example 10: The additive manufacturing system of any of examples 1 through 9, wherein the at least one topology sensor is configured to measure a topology of material added to the melt pool.

[0130] Example 11: A method for additive manufacturing includes receiving, by a computing device, topological data for a plurality of layers from one or more sensors of an additive manufacturing system, wherein the topological data is representative of a topology of a build surface of each layer of the plurality of layers; identifying, by the computing device, differences between the topological data and specification data representative of a set of tolerances of the build surface; controlling, by the computing device and based on a set of deposition parameters, an energy delivery device to deliver energy to the build surface to form a melt pool and a powder

delivery device to direct a powder stream toward the melt pool; and controlling, by the computing device, a machining device to machine the build surface based on the identified differences.
[0131] Example 12: The method of example 11, further comprising identifying at least one of the identified differences as a defect corresponding to a different density.
[0132] Example 13: The method of any of examples 11 and 12, wherein the plurality of layers includes a plurality of intermediate layers, and wherein at least one difference of the identified differences is identified in an intermediate layer of the plurality of intermediate layers.
[0133] Example 14: The method of example 13, further comprising generating as-deposited model data that includes the identified differences.
[0134] Example 15: The method of example 14, wherein controlling the machining device includes: generating machining data based on the as-deposited model data; and outputting the machining data to the machining device.
[0135] Example 16: The method of example 15, wherein the machining data includes: a toolpath; a first set of machining parameters for portions of the toolpath that do not include the identified differences; and a second set of machining parameters for portions of the toolpath that include the identified differences, wherein at least one machining parameter of the second set of machining parameters is different from the first set of machining parameters.
[0136] Example 17: The method of example 16, wherein the at least one machining parameter includes a change in material removal rate.
[0137] Example 18: The method of any of examples 13 through 17, further includes identifying the at least one difference for the intermediate layer; controlling the machining device to machine the build surface based on the at least one difference; and after machining the build surface, controlling the energy delivery device and the powder delivery device to deposit a subsequent layer on the intermediate layer.
[0138] Example 19: The method of any of examples 11 through 18, further includes evaluating whether at least one difference exceeds a machinability threshold; and in response to the at least one difference exceeding the machinability threshold, controlling the machining device to machine the build surface.
[0139] Example 20: The method of any of examples 11 through 19, wherein the at least one topology sensor is configured to measure a topology of material added to the melt pool.
[0140] Various examples have been described. These and other examples are within the scope of the following claims.

## Claims

**1**. An additive manufacturing system, comprising: an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool; a machining device configured to machine the build surface; one or more sensors comprising at least one topology sensor configured to generate topological data representative of a topology of the build surface; and a computing device configured to: receive the topological data from the at least one topology sensor for a plurality of layers; identify differences between the topological data and specification data representative of a set of tolerances of the build surface; control the energy delivery device and the powder delivery device to deposit the plurality of layers based on a set of deposition parameters; and control the machining device to machine the build surface based on the identified differences.
**2**. The additive manufacturing system of claim 1, wherein the computing device is configured to identify at least one of the identified differences as a defect corresponding to a different density.
**3**. The additive manufacturing system of claim 1, wherein the plurality of layers includes a plurality of intermediate layers, and wherein at least one difference of the identified differences is identified

in an intermediate layer of the plurality of intermediate layers.
**4**. The additive manufacturing system of claim 3, wherein the computing device is further configured to generate model data that includes the identified differences.
**5**. The additive manufacturing system of claim 4, wherein, to control the machining device, the computing device is configured to: generate machining data based on the model data; and output the machining data to the machining device.
**6**. The additive manufacturing system of claim 5, wherein the machining data includes: a toolpath; a first set of machining parameters for portions of the toolpath that do not include the identified differences; and a second set of machining parameters for portions of the toolpath that include the identified differences, wherein at least one machining parameter of the second set of machining parameters is different from the first set of machining parameters.
**7**. The additive manufacturing system of claim 6, wherein the at least one machining parameter includes a change in material removal rate.
**8**. The additive manufacturing system of claim 3, wherein the computing device is configured to, for an intermediate layer of the plurality of intermediate layers: identify the at least one difference for the intermediate layer; control the machining device to machine the build surface based on the at least one difference; and after machining the build surface, control the energy delivery device and the powder delivery device to deposit a subsequent layer on the intermediate layer.
**9**. The additive manufacturing system of claim 1, wherein the computing device is further configured to: evaluate whether at least one difference exceeds a machinability threshold; and in response to the at least one difference exceeding the machinability threshold, controlling the machining device to machine the build surface.
**10**. The additive manufacturing system of claim 1, wherein the at least one topology sensor is configured to measure a topology of material added to the melt pool.
**11**. A method for additive manufacturing, comprising: receiving, by a computing device, topological data for a plurality of layers from one or more sensors of an additive manufacturing system, wherein the topological data is representative of a topology of a build surface of each layer of the plurality of layers; identifying, by the computing device, differences between the topological data and specification data representative of a set of tolerances of the build surface; controlling, by the computing device and based on a set of deposition parameters, an energy delivery device to deliver energy to the build surface to form a melt pool and a powder delivery device to direct a powder stream toward the melt pool; and controlling, by the computing device, a machining device to machine the build surface based on the identified differences.
**12**. The method of claim 11, further comprising identifying at least one of the identified differences as a defect corresponding to a different density.
**13**. The method of claim 11, wherein the plurality of layers includes a plurality of intermediate layers, and wherein at least one difference of the identified differences is identified in an intermediate layer of the plurality of intermediate layers.
**14**. The method of claim 13, further comprising generating as-deposited model data that includes the identified differences.
**15**. The method of claim 14, wherein controlling the machining device includes: generating machining data based on the as-deposited model data; and outputting the machining data to the machining device.
**16**. The method of claim 15, wherein the machining data includes: a toolpath; a first set of machining parameters for portions of the toolpath that do not include the identified differences; and a second set of machining parameters for portions of the toolpath that include the identified differences, wherein at least one machining parameter of the second set of machining parameters is different from the first set of machining parameters.
**17**. The method of claim 16, wherein the at least one machining parameter includes a change in material removal rate.

**18**. The method of claim 13, further comprising, for an intermediate layer: identifying the at least one difference for the intermediate layer; controlling the machining device to machine the build surface based on the at least one difference; and after machining the build surface, controlling the energy delivery device and the powder delivery device to deposit a subsequent layer on the intermediate layer.

**19**. The method of claim 11, further comprising: evaluating whether at least one difference exceeds a machinability threshold; and in response to the at least one difference exceeding the machinability threshold, controlling the machining device to machine the build surface.

**20**. The method of claim 11, wherein the at least one topology sensor is configured to measure a topology of material added to the melt pool.